



GalaxyMaps

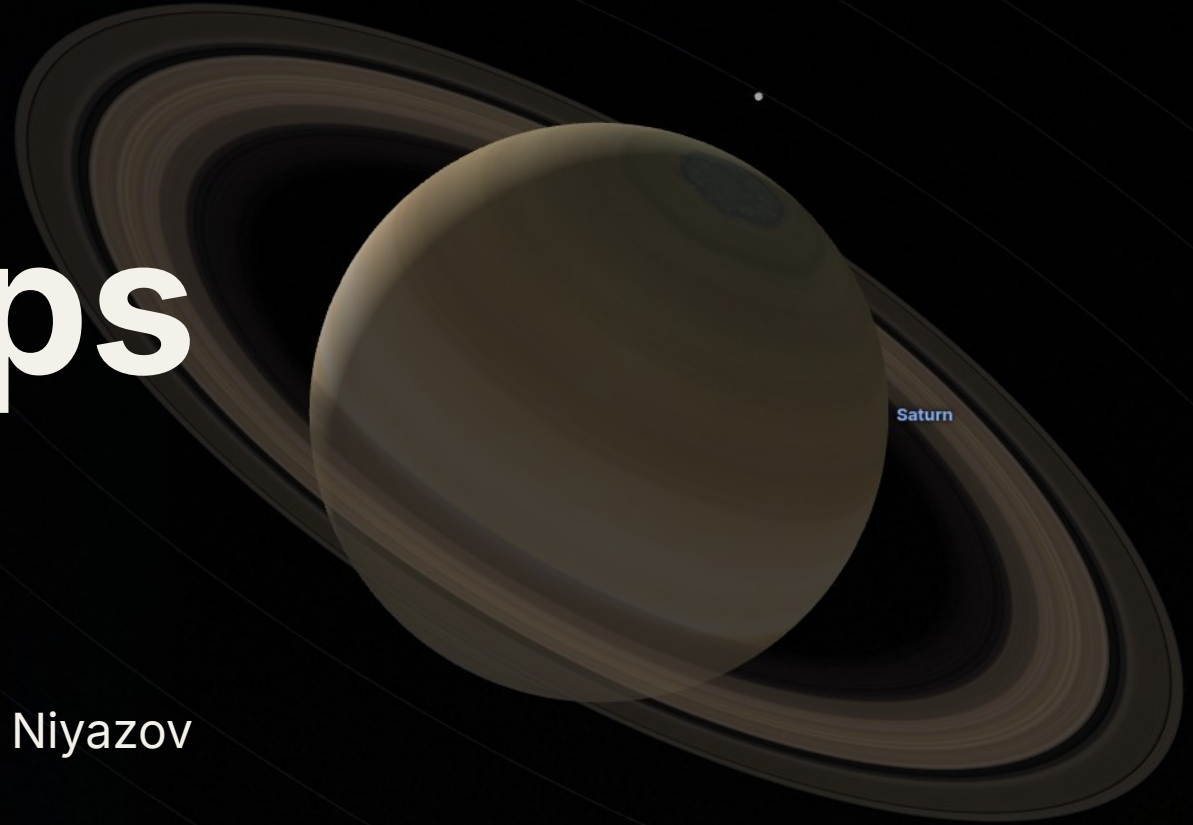
Navigate the universe.

TWRD

Teddy Lampert, Yijun Wang, Raj Iyer, Diyar Niyazov

BigRed//Hacks 2026
Major League Hacking

Built with Cursor
Voice navigation with Grok
SpaceX mission stories



Earth to Polaris

433 years

at the speed of light

Compare with Voyager 1

STRAIGHT-LINE CRUISE · VOYAGER 1

7.7 million years 433 light-years · 133 pc

Voyager 1 · 16.9 km/s · positions on 2026-10-04

Distance uncertainty ± 6.4 ly

GalaxyMaps
Directions across the universe

Directions

Earth ×
Polaris ×

+ Add stop

Travel mode
Light speed
299,792 km/s (c)

STRAIGHT-LINE CRUISE · LIGHT SPEED
433 years 433 light-years · 133 pc
Light speed · 299,792 km/s (c) · positions on 2026-10-04
about 5.4 human lifetimes (80 years each)
Distance uncertainty ± 6.4 ly

Begin journey
Camera travels the route · compressed visualization

0 seconds 10 s

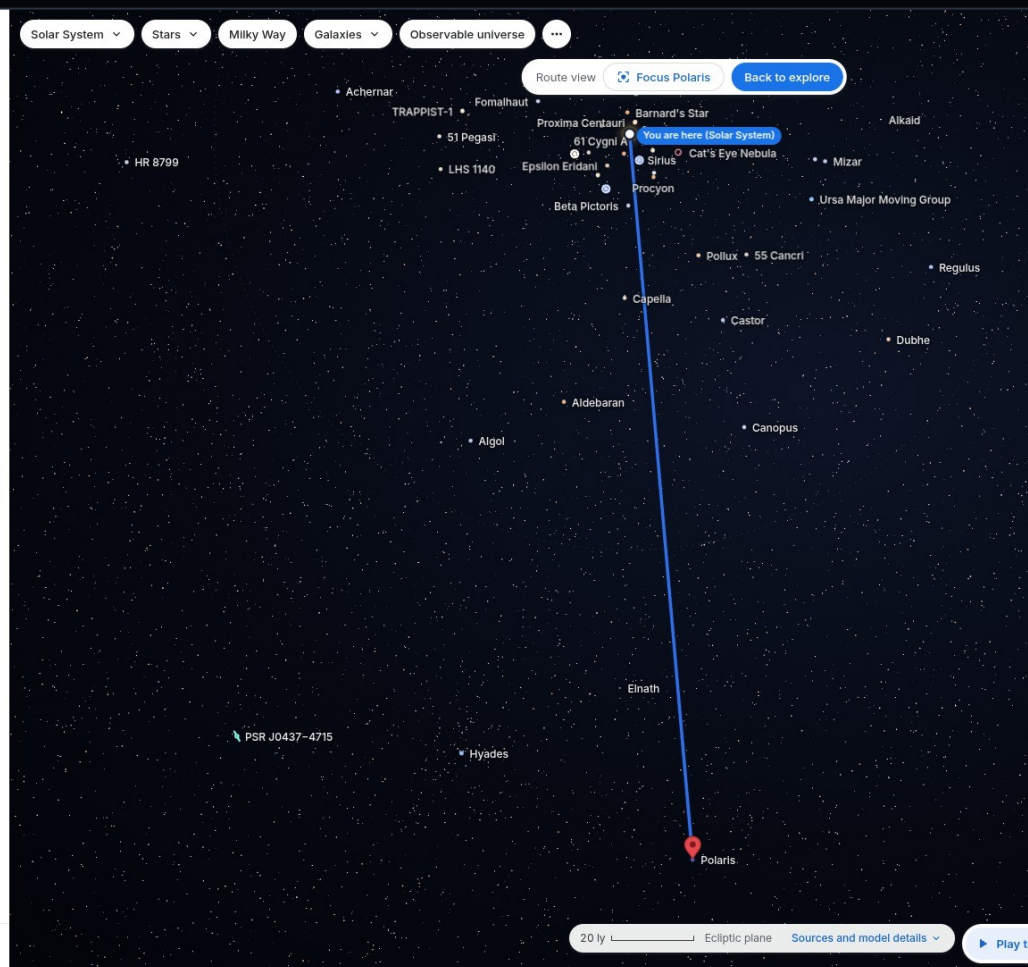
The preview has its own clock (10 s for the whole trip); Play time is paused while it runs.

► Sources and model details

Suggested stops · added distance

Vega +9.94 light-years (+2.3%)	On the way	+ Add
Capella +12.8 light-years (+3.0%)	On the way	+ Add
Van Maanen's Star +12.8 light-years (+3.0%)	On the way	+ Add

Mission Control



Approximate constant-speed comparison

"Show me Saturn."

Grok moves the map and answers questions along the way.



Apple Vision Pro · immersive view with the Grok mic HUD

Distances and travel times calculated by the app

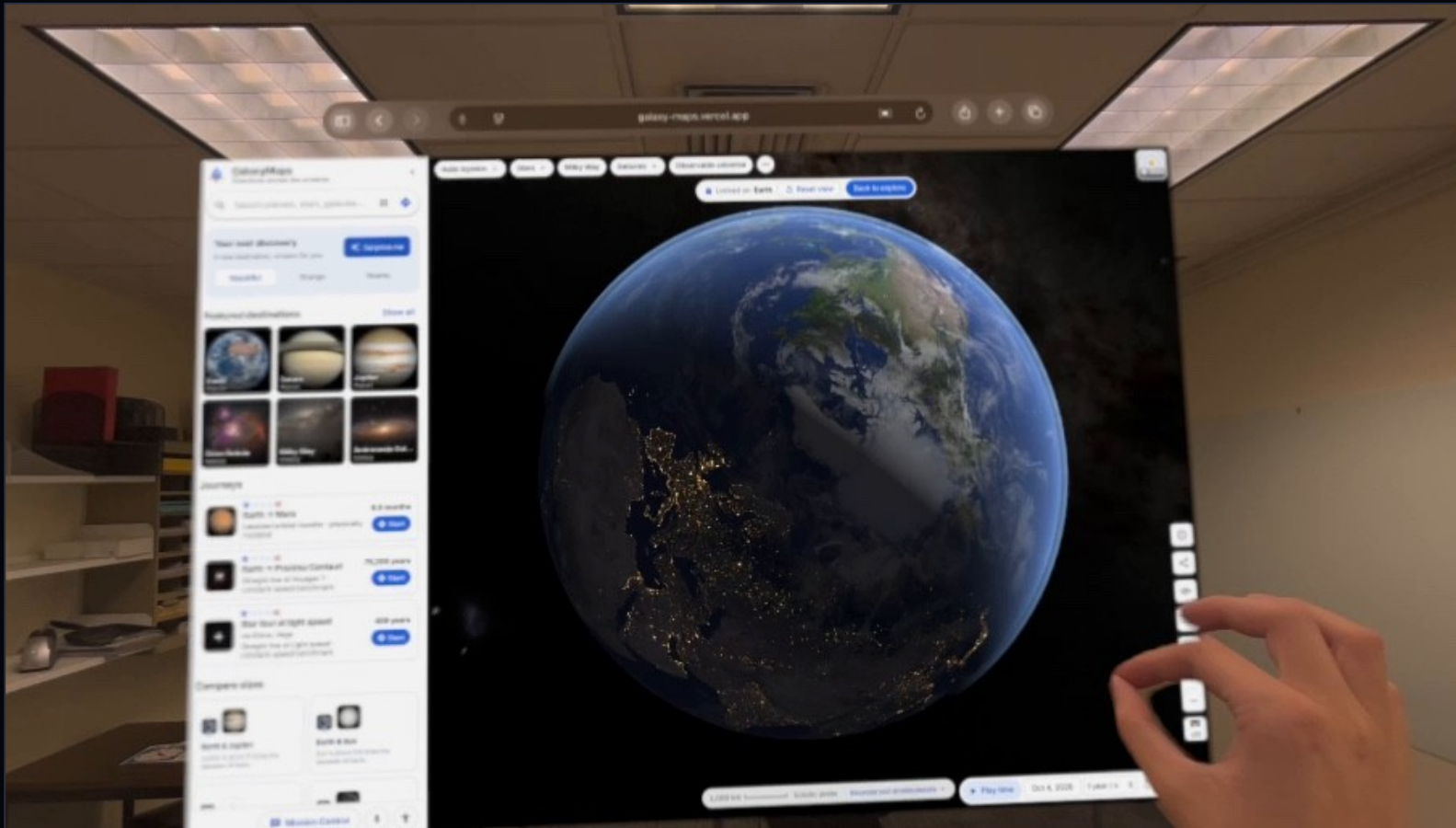
Space, from your classroom

A browser or a compatible
VR headset

Voice and keyboard navigation

Spoken descriptions

Larger text, high contrast,
reduced motion



Apple Vision Pro demo · GalaxyMaps in the spatial browser

Where would you go?



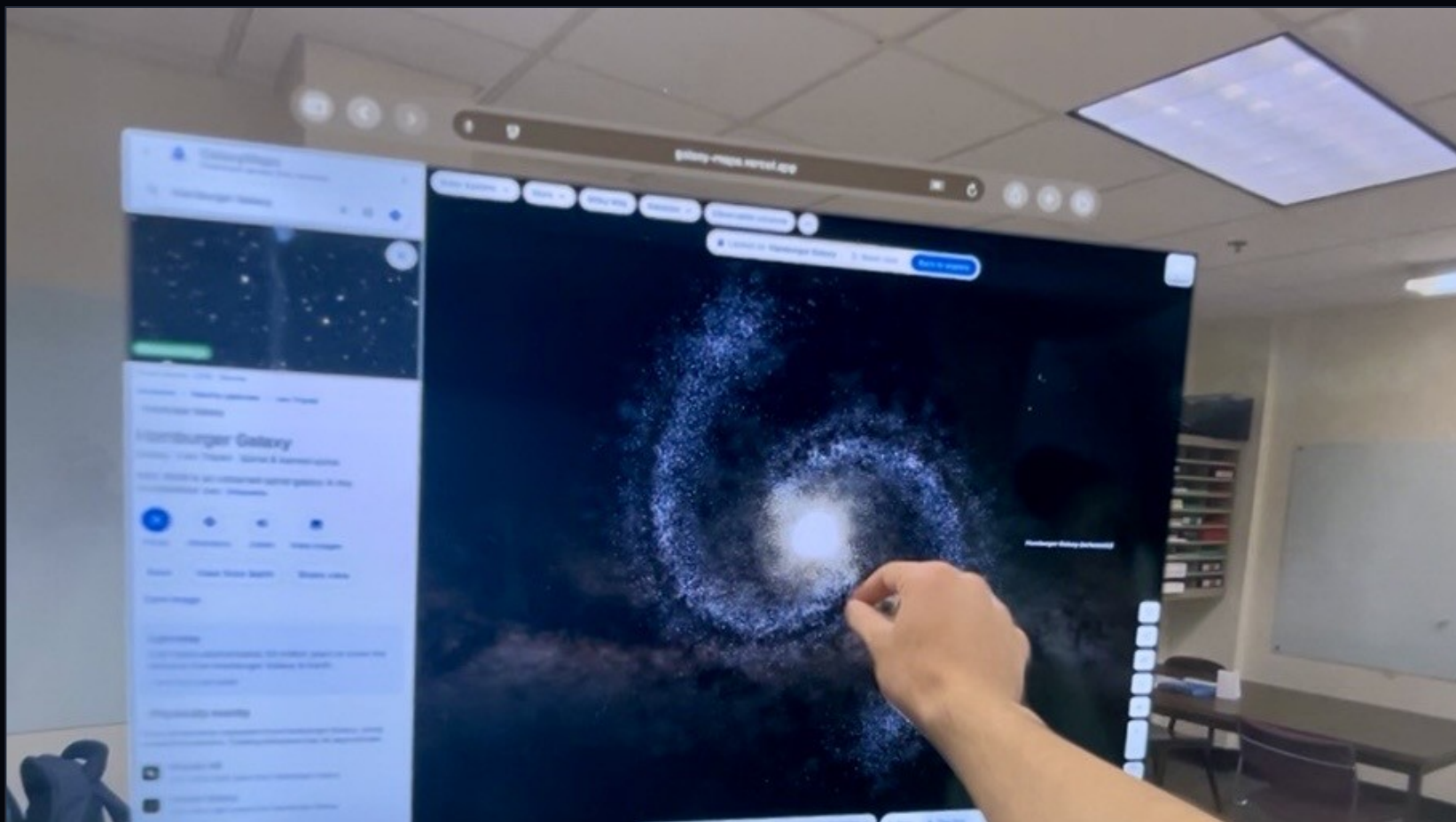
galaxies.wiki

TWRD



BACKUP

Exploring in Apple Vision Pro



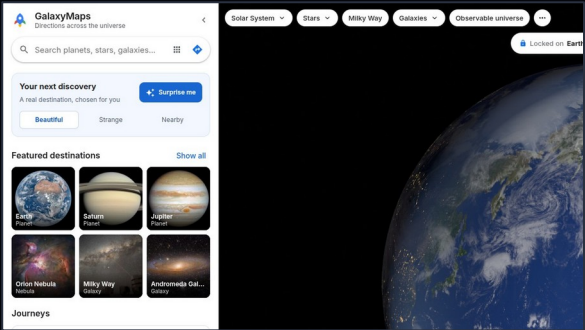
Galaxy views

Hand gestures in the room

The same map as the browser

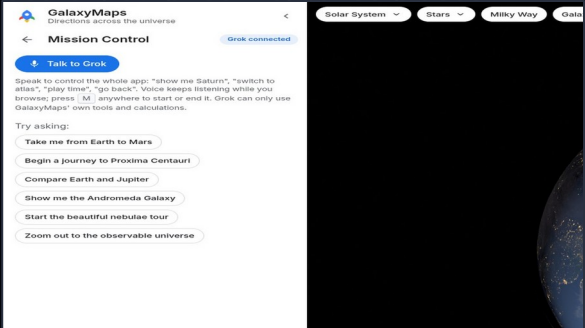
Apple Vision Pro · GalaxyMaps in the spatial browser

More of GalaxyMaps



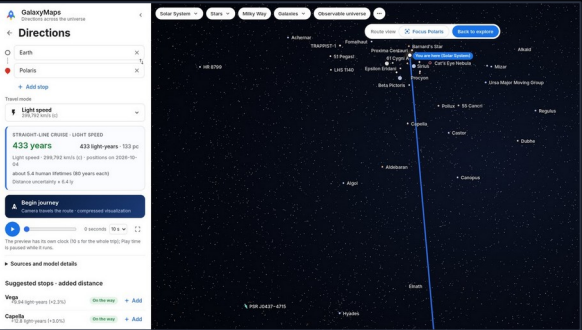
Explore

Search. Surprise me. saved places



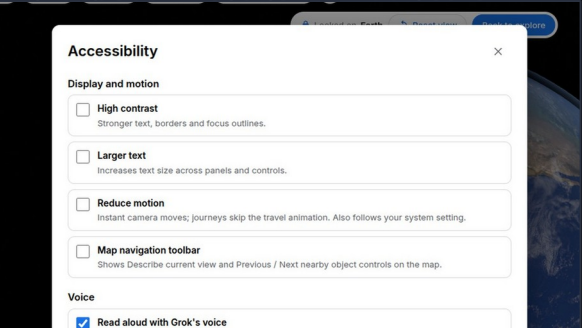
Grok

Voice or typed commands



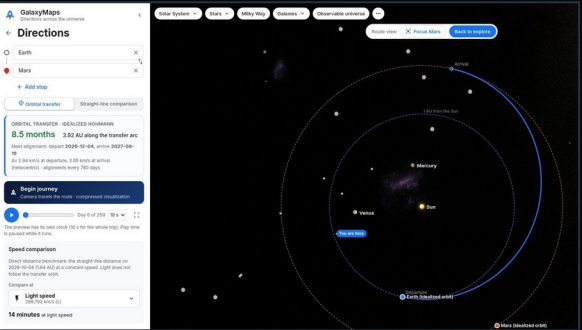
Earth to Polaris

Light speed · Vovader 1 · 433 years



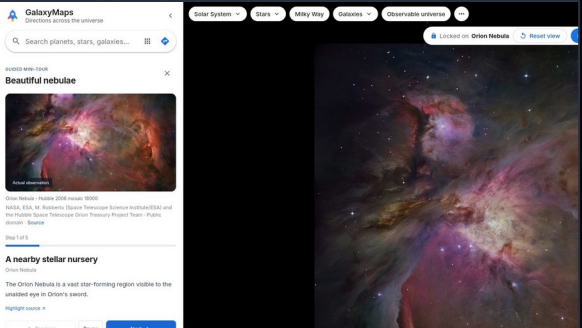
Accessibility

Keyboard, speech, contrast, motion



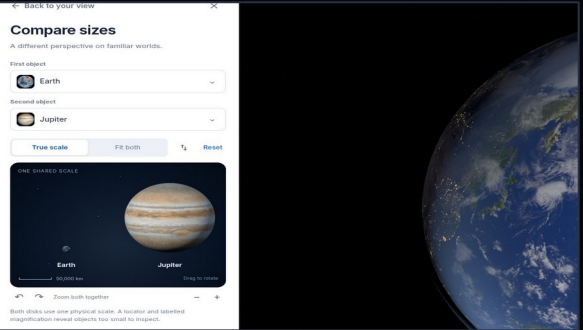
Earth to Mars

Idealized Hohmann · 8.5 months



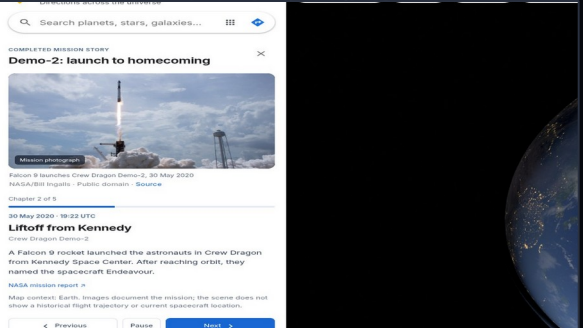
Guided tours

Beautiful nebulae · sourced photos



Compare sizes

True scale · Earth and Jupiter



Demo-2

Five NASA-sourced chapters